Stage 2: System design - high level

There are 3 main workflows

1) **User sets notification preferences** (User)
    a) User opens application
    b) User selects teams/games
    c) User toggles notifications
    d) App sends preferences to User Preference Service
    e) Preferences are stored
2) **Live event triggers notification** (Event driven)
    a) Sports Data Provider sends game event
    b) Event Ingestion Service receives event
    c) Event Processing Service processes event
    d) Event Processing queries User Preference Service for subscribers
    e) Game State Management updates game state
    f) Notification Service sends push notification
    g) Push Notification Service delivers notification to user device
3) **Use views live game information** (Read/view only)
    a) User opens app
    b) App requests game state
    c) Game State Management Service returns current score/status
    d) App displays game information

Component Features

Application (client)

- Create/manage user account
- Select team/games to follow
- Toggle alert types
- View notification history
- View live scores/updates

User Preference Service

- Store susbcriptions (maps teams/league/games to users)
- Returns subscribed users for an event
- User can update their preferences

Even Ingestion Service

- Receive events from sports provider
- Convert event data into proper format for system
- Forward events to Event Processing system
- If disconnected, try to connect back

Event Processing service

- Decides if an event should generate notifications
- Queries the USer Preference Service to get subscribers for a given event
- Updates game state
- Triggers the Notification Service

Game State Management Service

- Maintain most up to date score/status of games
- Stores recent/past event history
- Forwards game stat to application for user to see

Notification Service

- Build notification payload
- Send to Push notification service on user's device
- Write to the applications notification history

Java Public API skeleton

Models

- GameEvent
- UserPreferences
- GameState
- Notification

Services

- UserPreferenceService
    - void savePreferences(UserId user, UserPreferences prefs)
    - List<UserID> getSubscribers(GameEvent event)
    - UserPreferences getPreferences(UserId user)

- EventIngestionService
    - void onEvent(GameEvent event)

- EventProcessingService
    - void processEvent(GameEvent event)

- GameStateService
    - void updateGameState(GameEvent event)
    - GameState getGameState(GameId gameId)

- NotificationService
    - void sendNotification(List<UserId> users, Notification notification)
        - void sendNotification(UserId user, Notification notification)
    - List<Notification> getNotificationHistory(UserId user, int limit)

Diagram 3 - User Sets Notification Preferences BPMN

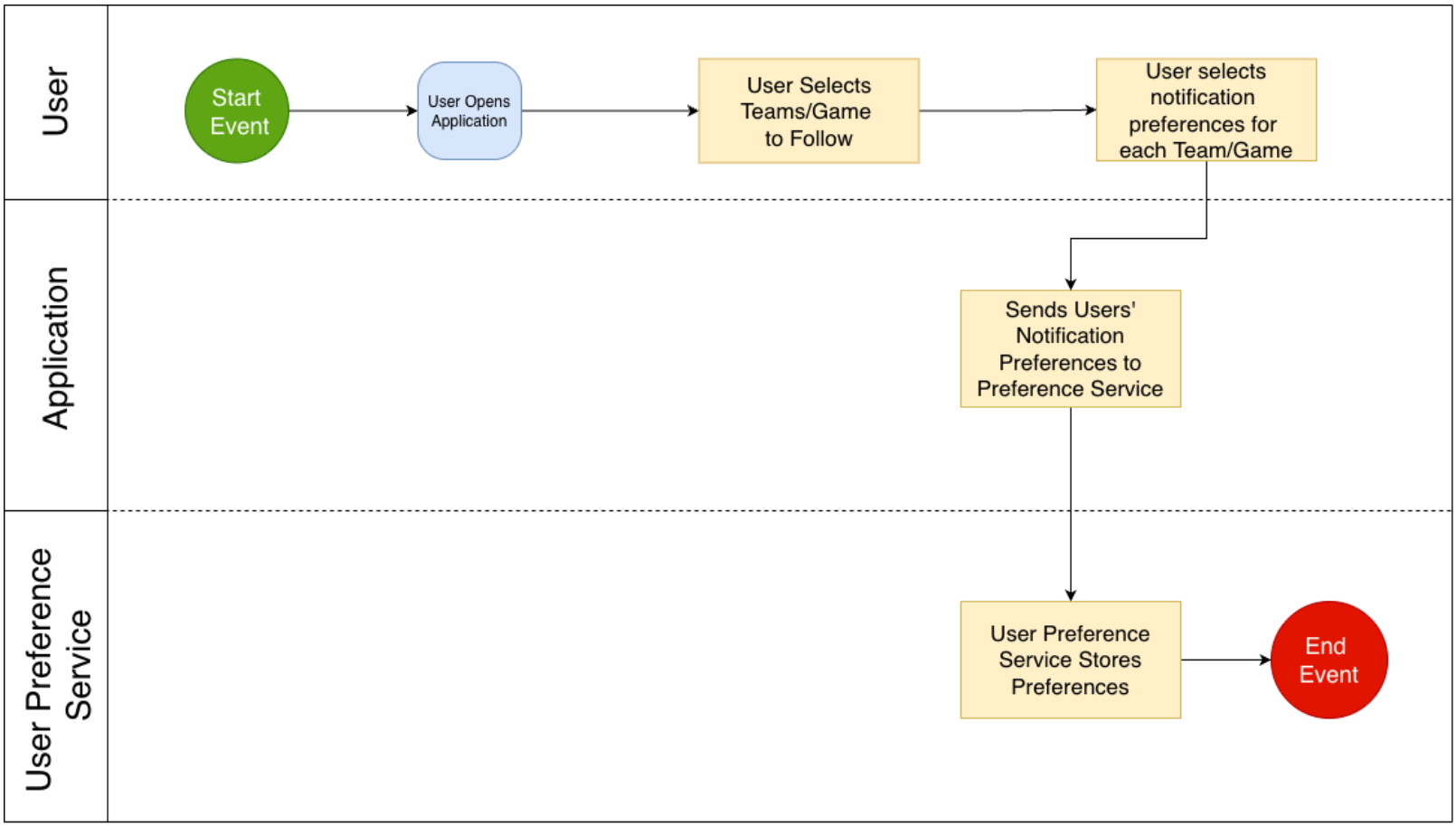


Diagram 4 - Live Event Notification BPMN

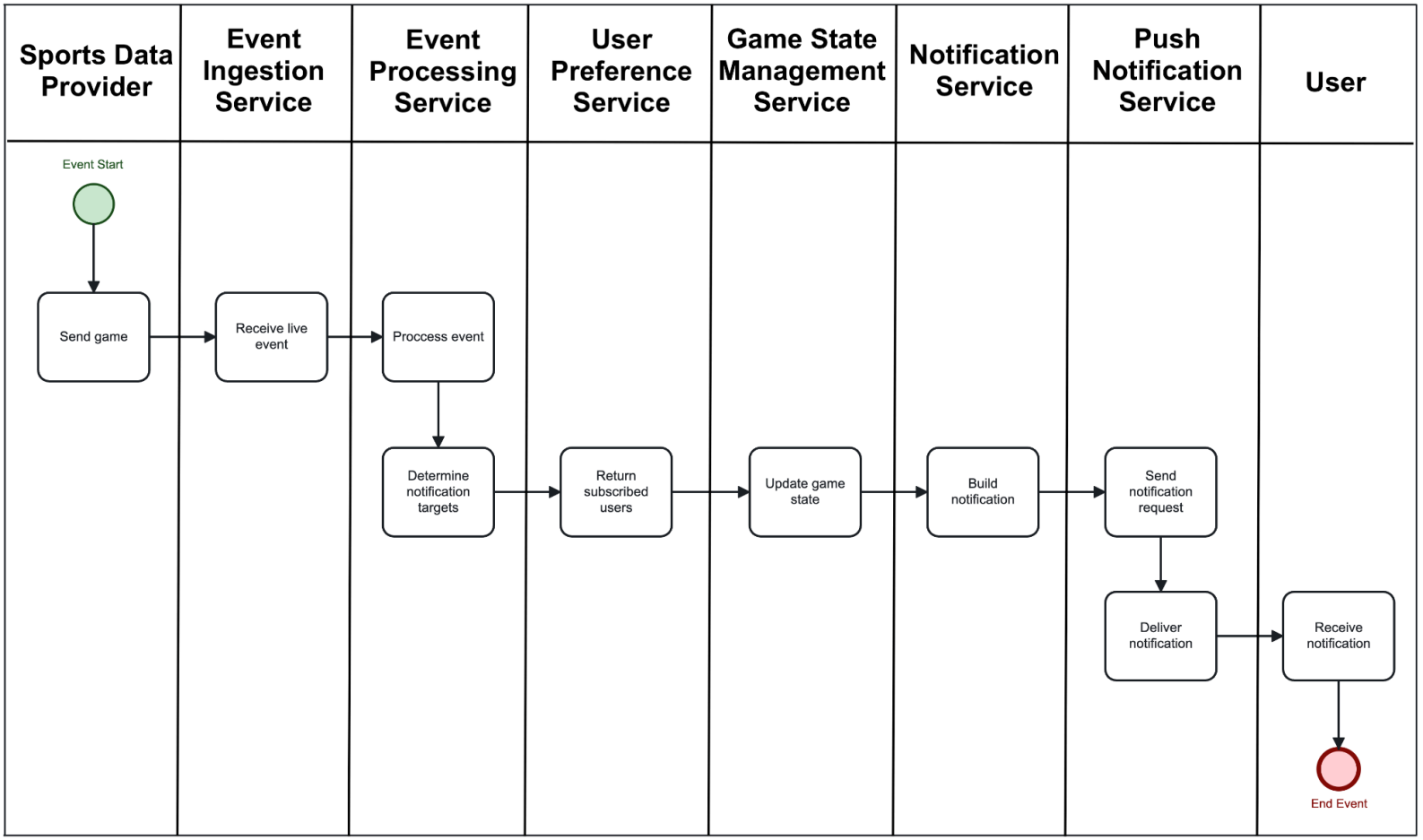


Diagram 5 - View Game State BPMN

